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Transforming abandoned mines into solar farms: a pathway to renewable energy development and sustainable land use

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Abstract

The demand for renewable energy is becoming increasingly apparent, but a significant challenge lies in the limited availability of land. To limit environmental impacts associated with new development in previously undisturbed lands, this study investigates the potential to convert abandoned mines in Florida and Pennsylvania into solar farms, aligning with federal and state-level clean energy goals and climate objectives. Thirteen abandoned mines in Florida and five in Pennsylvania were identified as suitable sites for solar farm development. These solar installations have the potential to generate up to 22% of Florida's and 0.017% of Pennsylvania's annual electricity needs. Employing various criteria such as acreage, slope, aspect, proximity to infrastructure, and insolation, the research assesses the feasibility of this conversion using advanced modeling techniques. The study was conducted in two phases: first, an in-depth spatial suitability assessment utilizing GIS tools (ArcGIS Pro) to evaluate potential mine sites based on specified criteria. Second, solar power potential and carbon emission reduction are estimated based on the use of two different solar panel setups with different installed capacity and of which one is static and the other one solar-tracking. This study underscores the importance of interdisciplinary approaches in addressing environmental and energy challenges by advancing our understanding of sustainable land use practices and contributing to the collective pursuit of clean energy objectives.

1. Introduction

The global push for renewable energy (RE) is accelerating; however, limited land availability presents a formidable challenge to large-scale implementation (NREL 2023). Projections indicate that the world will incorporate as much renewable power in the next five years as it did in the previous two decades, underscoring the urgency of this limitation (IEA 2022). In the United States, RE has seen significant expansion, with 20% of the country's electricity generation coming from sources such as solar, wind, and hydroelectric power in 2021 (DOE 2023). Among these technologies, wind and solar have seen the fastest growth in the last few decades. Wind energy surpassed hydropower as the nation's leading source of renewable electricity in 2019, while utility-scale solar capacity expanded from 61 gigawatts (GW) in 2021 to 71 GW in 2022, reflecting substantial infrastructure advancements (EIA 2023b, 2024b). While RE has progressed nationally, individual states face unique challenges. This study examines Florida and Pennsylvania (figure 1), two states with distinct energy landscapes yet shared goals in expanding RE. Both states have seen notable solar energy growth in the last decade; however, utility-scale solar installations require considerable land, complicating site selection (Ritchie 2022).

Florida's rapid urbanization further limits the availability of suitable sites for large-scale solar projects. The Agriculture 2040/2070 Report, a joint project between the University of Florida Center of Landscape Conservation Planning and 1000 Friends of Florida, projects a 23% population increase by 2040, further constraining available land for renewable projects (Agriculture 2024). Pennsylvania, in contrast, has rested on a legacy of coal and natural gas production but has been experiencing a shift toward RE adoption. The

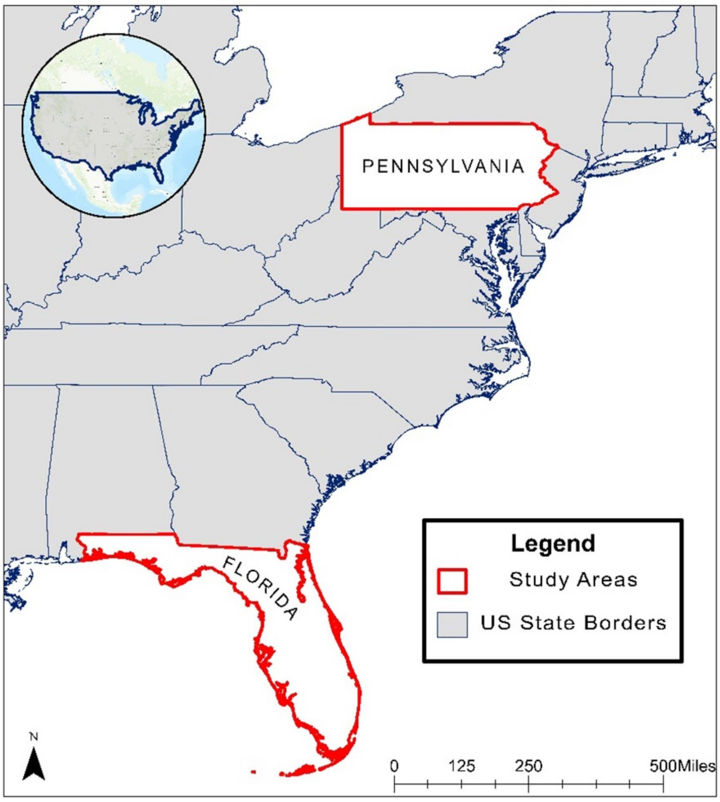

Figure 1. Map of study areas.

state is witnessing a growing interest in solar projects due to policy incentives and increasing awareness of RE benefits.

To confront the challenge of limited land availability, this study investigates the opportunity to utilize abandoned industrial sites for new utility-scale solar installations. Both Florida and Pennsylvania contain a significant number of abandoned mines, providing an opportunity to repurpose degraded land for clean energy generation. This research assesses the feasibility of repurposing abandoned mines in Florida and Pennsylvania into solar farms. The objectives of the study include:

- Identify suitable abandoned mines for solar farm development in Florida and Pennsylvania.
- Estimate solar energy production potential from selected suitable mine locations.
- Quantify CO_2 emissions reductions.

Overall, this research highlights repurposing abandoned mines for solar energy as an innovative land-use solution, demonstrating its potential to expand RE, reduce carbon emissions, and support sustainable development.

1.1. Incentives and programs

The Inflation Reduction Act (IRA), passed in 2022 under President Biden, addresses economic challenges post-COVID-19, focusing on inflation control and sustainable growth (EPA 2023). Among the many initiatives designed to jump-start the economy are also clean energy incentives, aimed to create a 50%–52% reduction in carbon emissions by 2030 compared to 2005 levels (The White House 2023). The IRA introduced the Powering Affordable Clean Energy (PACE) program, with $1 billion in loans for RE and storage projects in rural communities (US Department of Agriculture 2023). The PACE program supports RE developers and electric service providers, focusing on communities with populations under 20 000. It funds solar, wind, geothermal, biomass, hydropower, and energy storage projects, facilitating energy sales and advancing rural clean energy initiatives. Alongside this, the Bipartisan Infrastructure Law (BIL) strengthens US infrastructure, including $5 billion for Grid Resilience grants and support for Smart Grid initiatives to integrate renewable resources (DOE 2024). Together, the IRA and BIL showcase the US government's commitment to sustainable energy and infrastructure development.

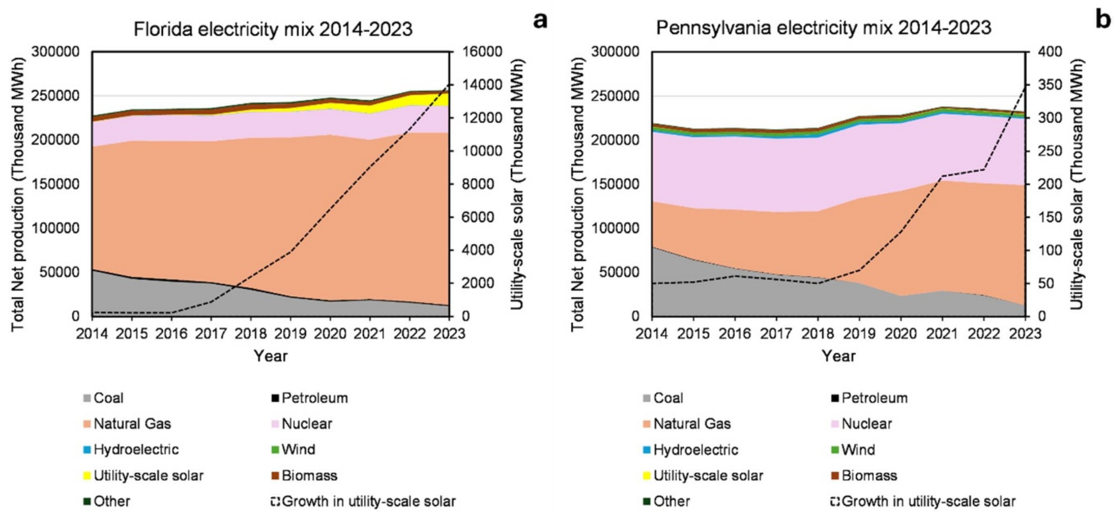

Figure 2. Electricity production mix in Florida (a) and Pennsylvania (b) for the years 2014–2023, based on data from EIA (2025). The dashed line shows the growth in solar energy during the same time period.

The Florida Brownfields Redevelopment Act incentivizes the cleanup and reuse of abandoned commercial and industrial sites to address public health and environmental risks (FDEP 2024). It offers Voluntary Cleanup Tax Credits (VCTC), cleanup liability protection, and expedited regulatory frameworks to foster economic revitalization and promote ecological equity (FDEP 2024). The act also supports job creation bonuses and tax refunds on building materials, bolstered by grant programs and collaborations with institutions like the Center for Brownfields Research and Redevelopment at the University of South Florida. Similarly, Pennsylvania's Brownfield Redevelopment provides a range of funding programs and incentives to support site assessment and cleanup efforts (PADEP 2024). The Department of Environmental Protection (DEP) and the Department of Community and Economic Development (DCED) offer funding programs, including the Industrial Sites Reuse Program, Pennsylvania First Program, and Business in Our Sites initiative, while the PENNVEST Brownfields Remediation Loan Program provides financial support for remediation (PADEP 2024). Federal agencies, such as the Environmental Protection Agency (EPA) and the US Department of Agriculture (USDA), also contribute direct funding and support for brownfield revitalization projects (USDA 2017, EPA 2024). DEP serves as a primary point of contact for economic development agencies, businesses, and government officials, providing information, training, and unique assistance to facilitate successful brownfield redevelopment projects.

1.2. State electricity mix for Florida and Pennsylvania

Figure 2 shows the electricity mix for Florida and Pennsylvania for the years 2014–2023. Over these ten years the usage of coal has diminished significantly in both states, and it has been replaced mainly by natural gas, but to some extent also renewable energies. Figure 2 also shows the growth of solar energy during the same time period. Solar power is the RE source that has increased the most in both states, approximately seven-fold in Pennsylvania, while Florida boasts with 58 times more solar production in 2023 than ten years earlier. This is likely a reflection of the fact that Florida has very few energy resources but relies on importing fuels from other states, but also its geographic location. Pennsylvania on the other hand has significant fossil fuel resources and has seen less of a push to develop renewable energies. As of 2023, renewable energies contributed approximately 3% to Pennsylvania's electricity mix, and 7% to Florida's (EIA 2025).

1.3. RE policies

Florida and Pennsylvania have taken significant steps toward RE and climate goals over the past few decades. Florida's efforts began in the early 2000s with initiatives promoting RE, building on the foundation set by the Florida Energy Efficiency and Conservation Act (FEECA) of 1980 (DSIRE 2019). Similarly, Pennsylvania's RE journey started with the 2004 Alternative Energy Portfolio Standard (AEPS) (Pennsylvania General Assembly 2004). Followed by RPS adoption in 2007 (PAPUC 2022). Efforts focused on solar and wind energy, energy efficiency, and the Future Solar Plan of 2018 (DEP 2018), and were further strengthened by the Clean Energy Program Plan of 2020 (DEP 2020), all contributing to the state's goal of reducing greenhouse gas emissions by 80% by 2050 (DEP 2021). Both states highlight legislative efforts to promote clean energy and environmental sustainability.

1.4. The challenge of land availability

A limitation of renewable energies is their need for significant amounts of land. Considering the full life-cycle of utility-scale powerplants, solar panels trails behind most traditional energy sources in its efficiency measured as land use per unit of electricity (Ritchie 2022). One approach to this challenge is to re-purpose already developed land, which has lost its original objective, to utility scale solar energy. This method is often referred to a 'brown field development' (Greenberg *et al* 2001).

While 'greenfield projects', that is development or construction on pristine lands, require meticulous scrutiny of permits, regulations, and environmental impacts, and risk causing adverse environmental impacts (Abrahams *et al* 2015), brownfield redevelopment presents innovative solutions for revitalizing these areas (Greenberg *et al* 2001). Brownfields, estimated at 450 000 sites across 15 million acres in the US, offer promising avenues for leveraging existing infrastructure near urban hubs (EPA 2024). By repurposing these sites, communities can breathe new life into formerly neglected areas, potentially bolstering tax revenues and fostering employment. A study of six landfill redevelopment projects in New Jersey by Sadat Associates, Inc. demonstrates the potential of transforming these sites into valuable assets, including a community college, regional mall, offices, golf courses, and housing, covering 378 acres of waste (Wiley and Assadi 2002). The study emphasizes financial incentives, economic viability, and community relations, illustrating the feasibility and benefits of brownfield redevelopment. This study underscores the importance of considering factors like contamination levels, regulatory policies, and community engagement in revitalizing these sites while balancing the need for greenfield development. While traditional redevelopment focuses on economic benefits, there is a growing recognition of brownfields' potential for greening urban environments. Research from De Sousa (2003), aimed to understand the challenges and benefits of converting contaminated brownfields into green spaces like parks and trails and, specifically, looking at the experience of the city of Toronto (Canada), which offers valuable insights for cities seeking to improve their urban quality of life through brownfield planning. However, the concept of using old mine sites, which have been abandoned, is not well studied. Contemporary energy discussions focus on exploring how to implement RE sources in various industries, including mining (Pollack 2017). The Nature Conservancy reported the benefits of solar energy development on former mine sites in 2023, emphasizing its alignment with efforts to promote RE adoption and mitigate environmental impacts. Their findings underscored the significance of repurposing abandoned mine lands (AMLs) for solar energy production as a sustainable solution to address land scarcity and advance clean energy goals (The Nature Conservancy 2023). Additionally, research conducted by Jones (2023) examined mining properties in the state of Nevada for utility-scale solar development, while Nasirov and Agostini's (2018) explored integrating solar energy with operational mines in Chile. Both studies highlighted the potential of solar energy development on AMLs to mitigate environmental impacts and promote RE adoption. Nasirov and Agostini 's (2018) research particularly emphasized that since mining operations predominantly drive Chile's energy consumption, solar resources could adequately satisfy the mining industry's demand, thus offering a sustainable energy solution. Amazon transformed a coal mine in Maryland into the state's largest solar project named Amazon Solar Farm Maryland, demonstrating the feasibility and scalability of solar energy development on former mine sites (Olick 2023). Similarly, The Nature Conservancy repurposed six old coal mines into utility-scale solar farms, creating the Cumberland Forest Project in Virginia, showcasing the practical application of solar energy to reclaim and revitalize degraded lands (The Nature Conservancy 2020). The backdrop of evolving RE policies in Pennsylvania and Florida serves as a contextual foundation, emphasizing the symbiotic relationship between regional legislative initiatives and the transformative potential of repurposing abandoned mines for solar energy production. Furthermore, research by the EPA (2011) report provides insights into the regulatory landscape and policy implications of integrating RE projects on reclaimed mine lands, further emphasizing the importance of aligning state-level initiatives with sustainable land use practices (EPA 2011). The paper by Pai *et al* (2020) underscores the techno-economic advantages of solar energy over wind energy in replacing coal mining jobs, offering additional support for the feasibility and benefits of solar development on reclaimed mine lands. This collective body of research underscores the multifaceted benefits and potential of repurposing AMLs for solar energy production, highlighting a promising pathway towards RE expansion and sustainable land use.

1.5. Mining history

Florida's mining history is deeply tied to phosphate rock extraction, which began in 1888 and became a major industry, particularly in the Bone Valley region (Matson 1915, Pittman 1990). Phosphate mining spans over 450 000 acres and contributes 80% of US and 25% of global production (FIPR 2020). However, fluctuating demand and environmental concerns led to many mine closures, leaving ecological issues behind (Duan *et al* 2021). In response, Florida enacted reclamation standards requiring the restoration of mined lands post-1975, focusing on water quality, revegetation, and wetland restoration (Beavers *et al* 2013, FDEP 2023). Today, the state oversees projects like the 75 MW Fort Green Power Plant, repurposing former mine

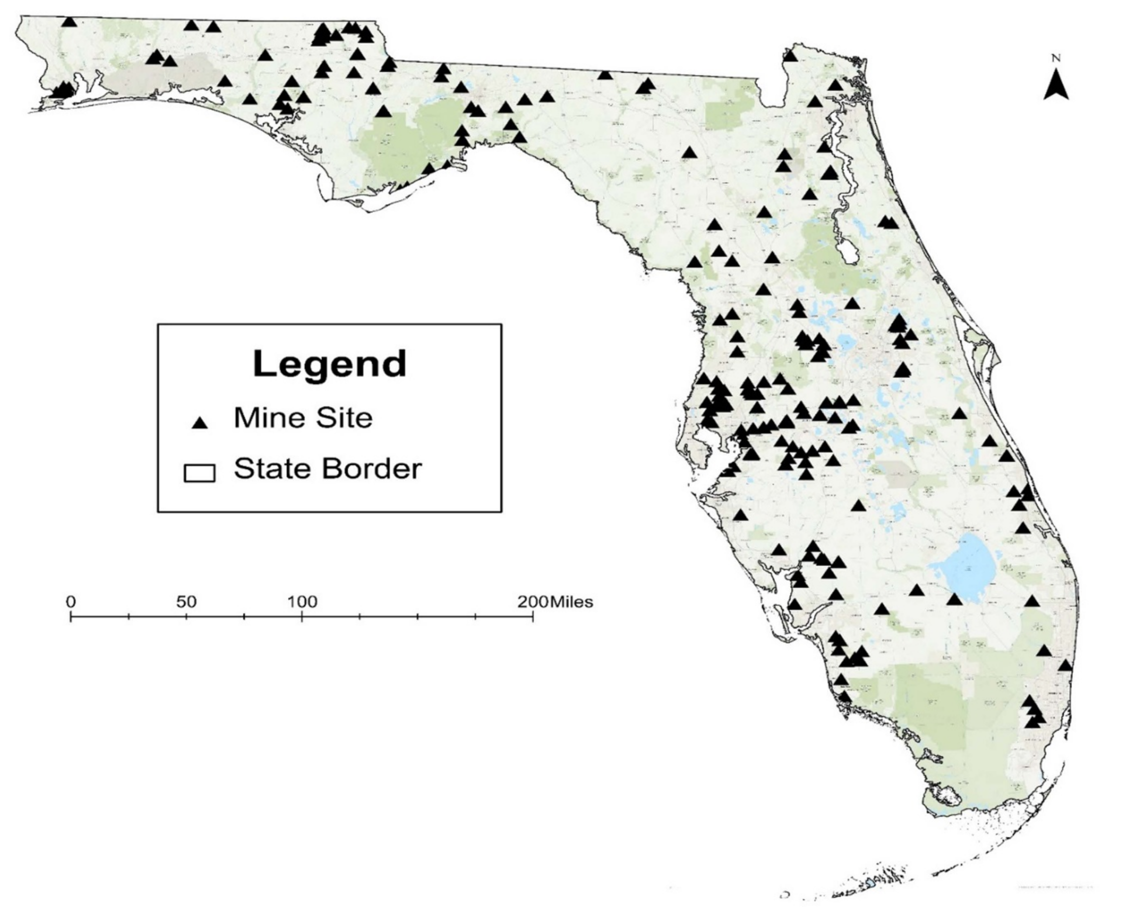

Figure 3. 2021 abandoned mine lands in Florida (phosphate and non-phosphate mines) data source (FDEP 2022).

sites for solar energy (Mosaic 2021). Additionally, Florida's non-phosphate mines, including those for zircon and titanium, offer potential for solar development, helping diversify the state's economy, increase energy independence, and promote sustainable land use (FDEP 2022). Figures 3 and 4 are maps of abandoned mines across the states of Florida and Pennsylvania.

Pennsylvania's mining history dates back to the mid-1700s, when its coal industry played a key role in fueling the nation's industrial sector, peaking at 276 million tons of coal in 1918 (PADEP 2023 Coal remained vital for electricity generation into the 20th century, but the environmental impacts of coal mining were largely neglected, leading to over 250 000 acres of AML (DEP 2023). These AMLs pose environmental hazards, such as acid mine drainage, and create safety risks (Liu *et al* 2012). Pennsylvania has led efforts to establish mining regulations since the 1960s, but the state still faces over $1 billion in health and safety problems related to AMLs (2023). Pennsylvania is responsible for one-third of the country's AML issues, with sites spread across 43 counties (Dixon and Bilbrey 2015). Refer to figure 2 for a detailed map of abandoned coal mines across Pennsylvania in 2022. Despite these challenges, the solar industry offers hope for the future, providing employment opportunities for former miners and contributing to economic diversification and sustainable land use practices.

2. Materials and methodology

The guidelines and recommendations from the National Renewable Energy Laboratory (NREL) and the Solar Energy Industries Association (SEIA) provide extensive information on the implementation of utility-scale solar farms, which serves as the basis for this study. These criteria include land area, south-facing angle, slope, land cover, and proximity to infrastructure, which are pivotal in evaluating site suitability (Ong *et al* 2013, SEIA 2024). Using R Studio and ArcGIS Pro version 3.0.1, the analysis starts with a focused evaluation of mine site suitability. To view a flowchart that illustrates this process, please refer to figure 5, which shows the main steps of the spatial suitability assessment. Site suitability is measured by overlaying variables including mine acreage, south-facing angle, slope, land cover, and proximity to infrastructure.

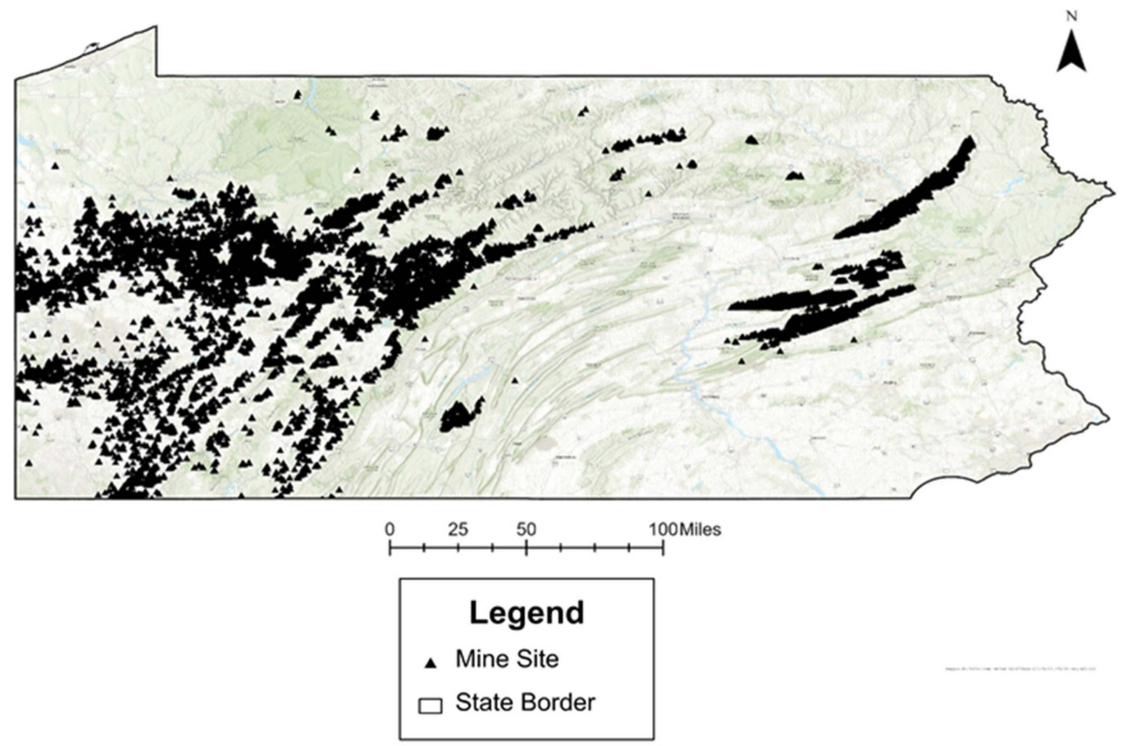

Figure 4. 2022 abandoned coal mine lands in Pennsylvania data source: (PDEP 2023).

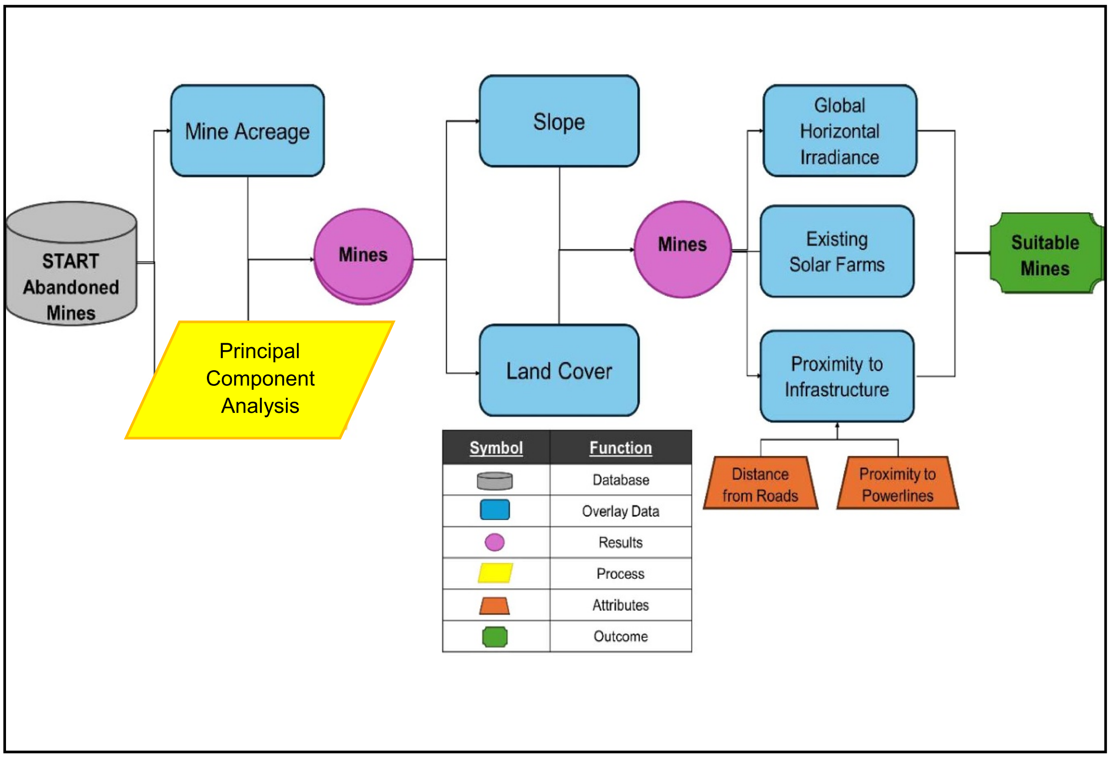

Figure 5. Spatial suitability assessment flowchart.

Table 1. Datasets for spatial suitability assessment.

Dataset	Data type	Resolution	Criteria	Source
Existing solar farms	Vector (point)	—	Avoid overlap with existing solar farms	US Energy Information Administration (2024a)
FL abandoned phosphate mines	Vector (polygon)	—	Minimum 5 acre size	Florida Department of Environmental Protection (2019)
FL Abandoned non-phosphate mines	Vector (polygon)	—	Minimum 5 acre size	Florida Department of Environmental Protection (2024)
PA abandoned mines	Vector (polygon)	—	Minimum 5 acre size	Pennsylvania Department of Environmental Protection (2023)
National Elevation Dataset	Raster	100 m	Slope less than 5 degrees	National Atlas of the United States (2012)
Global horizontal irradiance	Raster	1 arc-sec	Greater than 3.0 kWh m^{-2} d^{-1}	Global Solar Atlas (2019)
US Electric Power Lines	Vector (line)	—	Less than 5 miles from power lines	Homeland Infrastructure Foundation-Level Data (2022)
North American land cover	Raster	30 m	Includes barren land; Excludes forests, wetlands, and urban areas	North American Land Change Monitoring System (2024)
Protected areas database	Vector (polygon)		Exclude sites within protected areas	U.S Geological Survey (2022)

Table 1 contains further information about the datasets included in the geospatial evaluation. Global horizontal irradiance (GHI) data and proximity to existing solar farms were also included in the assessment, ensuring a comprehensive evaluation of solar potential and integration with existing infrastructure for optimized project planning and success. This strategy guarantees that the strict selection standards are met, which ultimately leads to the identification of the best sites for solar farms on former mine lands. The PV Watts® Calculator from NREL is then used to determine the potential solar power generation for each site, and the state-specific energy mix is used to estimate the decrease in CO_2 emissions.

Florida's spatial suitability assessment for solar farm development began with a dataset of 215 potential locations, refined using PCA and a mine acreage overlay to identify promising sites. Additional overlays of NED slope, land cover, GHI, existing solar farms, and proximity to infrastructure further narrowed the selection. Similarly, Pennsylvania's assessment started with 17 573 abandoned mines, reduced using PCA to focus on south-facing angles, and further refined through overlays of topographical, environmental, and infrastructure criteria to determine suitable locations.

2.1. Spatial suitability assessment

2.1.1. Mine acreage

Filtering mines based on acreage criteria involves employing fundamental GIS techniques. Through an examination of the attribute table within any GIS software, focus is directed toward the column representing the area of each mine. By sorting or filtering this column, mines surpassing the minimum five-acre threshold are selected (SEIA 2024). Sites boasting ample land not only accommodate larger solar arrays, thereby enhancing energy outputs and furthering RE objectives, but also facilitate the incorporation of necessary infrastructure, such as solar panels, inverters, and access roads (Nix 2024). Furthermore, excluding mines with areas less than five acres or 0.02 km^2 ensures that selected sites possess sufficient space for optimal solar farm development, including appropriate spacing between arrays to maximize sunlight exposure, minimize shading, runoff access, and maintenance access (Yavari *et al* 2022). Additionally, the NREL highlights that utility-scale PV installations require an average of seven acres per MW of capacity. While five acres is on the lower end of this range, it was chosen as a practical minimum to ensure selected sites can still support necessary infrastructure and operations (Ong *et al* 2013). By prioritizing sites with adequate size, this research underscores the feasibility and efficacy of solar farm projects on AMLs, thereby enhancing their potential to contribute to sustainable energy initiatives significantly.

2.1.2. Principal component analysis (PCA)

PCA is a vital technique for determining the major axis or ideal orientation of polygons, which are commonly used to represent land parcels or geographic areas (Chi 2021). It was applied in the analysis datasets pertaining to the study of abandoned mines in both Pennsylvania and Florida. Specifically, PCA was employed on raster datasets representing the topographic features of the mine sites (FL Abandoned Phosphate Mines, FL Abandoned Non-Phosphate Mines, and PA Abandoned Mines). Initially, PCA standardizes raster datasets representing mine site topographic features like elevation, slope, and aspect to ensure consistency in scale. Subsequently, PCA transforms these datasets into PCs, which include eigenvectors defining optimal orientations and a derived south-facing angle for each site. This angle is crucial for identifying the best positioning of solar panels to optimize solar energy utilization on AMLs. Then the PCA operates by transforming the original polygon coordinates into linearly uncorrelated variables, allowing us to determine the predominant direction or major axis. In context, PCA is particularly useful because it helps capture the variability in the data and extract meaningful patterns that might not be apparent in the original coordinate system. This is vital for this project, where the aim is to identify the optimal orientation of polygons, such as land parcels or geographic areas, for solar energy capture (Lendave 2021). Moreover, PCA ensures consistency in the analysis by adjusting negative angles to ensure measurement clockwise from the positive x-axis. This standardized approach allows for a consistent interpretation of the results and facilitates comparisons across different datasets or geographic regions. Ultimately, the calculated angle derived from PCA provides valuable input for determining the ideal positioning of solar panels on selected sites, thereby contributing to the overarching objective of optimizing solar energy utilization on AMLs.

2.1.3. Slope

The slope analysis employed in this study is critical for identifying suitable mine sites by considering slopes of less than five degrees and prioritizing flat terrains conducive to efficient solar infrastructure installation (Kereush and Perovych 2017). Leveraging the National Elevation Dataset (NED) with its 100 m resolution, the method calculates the slope for each south-facing polygon using a custom function (Gesch *et al* 2002). This function computes the slope based on the coordinates of the polygon vertices, determining the vertical-to-horizontal extent ratio. The calculated slopes are then compared to a predefined threshold, equivalent to the tangent of five degrees converted to radians, to filter out polygons with five degrees or more slopes (SEIA 2024). By identifying topographically suitable circumstances for solar energy development on specific mine sites, the analysis efficiently contributes to maximizing energy output from solar farms by concentrating on mines that fulfill the stated slope criteria of five degrees or less.

2.1.4. Land cover

Assessing land availability and cover is crucial for identifying suitable sites for solar farm installations, considering ecological and environmental factors. Optimal land cover, such as open fields or degraded lands, provides favorable conditions for solar arrays, enabling efficient energy generation (Kereush and Perovych 2017). In this analysis, land covers suitable for solar development include barren land and areas with minimal vegetation (e.g. tropical or sub-tropical grassland). Conversely, dense vegetation, water bodies, wetlands, urban and built-up areas, and forests were excluded from consideration due to their unsuitability for solar farm installations.

Although the land cover is limited to abandoned mines, there is variability due to differences in post-closure conditions. Some mines remain barren or minimally vegetated, while others have experienced natural regeneration or reclamation. The dataset did not include information on reclamation stages, but this variability informed the exclusion of unsuitable land covers like dense vegetation and wetlands.

It is worth noting that while the dataset used for land cover assessment provides valuable information, it did not include data on protected areas. To address this limitation and ensure a comprehensive analysis, the Protected Areas Database (PAD-US) 3.0 Data was utilized to identify and exclude mine sites that were in protected areas from consideration (USGS 2022). Overlapping protected areas with AMLs provides essential contextual information and helps address potential inquiries regarding environmental conservation efforts in the study area.

2.1.5. Proximity to infrastructure

Building infrastructure for a new power plant is often a significant part of the initial investment. Considering existing infrastructure such as roads and powerlines is therefore paramount in evaluating site suitability for solar farm installations on AMLs. Overlaying the distance from roads and assessing proximity to powerlines helps ensure efficient connectivity and reduces costs associated with infrastructure development (Meirzwaik and Calka 2017). Sites closer to existing roads require less investment in building new access routes and facilitating transportation of equipment and materials during the construction and maintenance phases. Additionally, proximity to high-voltage powerlines minimizes transmission losses and infrastructure expenses by enabling direct connection to the grid, thus enhancing the economic viability of solar projects (EPA 2011). Integrating these factors into site suitability assessments ensures optimal infrastructure utilization, streamlines project development and contributes to solar farm initiatives' overall success and sustainability on AMLs.

2.1.6. Existing solar farms

Enhancing the strategic planning and effectiveness of solar energy projects is the goal of an existing solar farm analysis, especially when it comes to choosing appropriate locations for future installations. The present study entails overlaying selected mine sites with existing solar farms to optimize spatial resources and prevent superfluous development. The research aims to ensure smooth integration of new solar installations with existing infrastructure, optimizing land use efficiency, preventing conflicts or competition between projects, and alleviating potential strain on the electric grid (NREL 2010), achieved through strategically spacing new solar farms to avoid overlap with existing installations. By leveraging insights from existing solar farm locations, this approach ensures a cohesive and coordinated expansion of solar energy infrastructure, maximizing overall energy generation potential and contributing to a more sustainable energy landscape. Additionally, considering factors such as grid capacity and transmission capabilities further enhances the resilience and reliability of solar energy systems, ensuring seamless integration into the broader energy infrastructure.

2.1.7. GHI

In this analysis, the average annual GHI values for Florida and Pennsylvania were sourced from the Global Solar Atlas. GHI is defined as the total amount of solar radiation received per unit area (kWh m^{-2}) on a horizontal surface and is a crucial metric for assessing solar energy potential (Sandia National Laboratories 2024). This platform provides comprehensive solar resource data, including GHI, which is derived from satellite observations, meteorological models, and ground-based measurements. The average annual GHI for Florida is approximately 4.5 kWh m^{-2} d^{-1}, while for Pennsylvania, it is approximately 3.75 kWh m^{-2} d^{-1} (Global Solar Atlas 2019). By utilizing these values, variations in solar radiation throughout the year are considered, providing a standardized metric for evaluating solar energy viability across diverse geographical regions within each state. This approach ensures a uniform basis for assessing solar energy potential, facilitating consistent analysis and comparison across the study area.

GHI maps were generated using data from the Global Solar Atlas to visually represent the spatial distribution of GHI across Florida and Pennsylvania. Figure 6 illustrates the distribution of GHI across Florida, highlighting areas with higher solar radiation intensity, while 7 depicts the GHI distribution in Pennsylvania. These maps provide valuable insights into the geographic variation of solar resources, aiding in identifying optimal locations for solar energy development within each state (Global Solar Atlas 2019). Understanding these variations is essential for informed decision-making regarding site selection for solar farm development on AMLs.

2.2. Power production potential and carbon emission reduction analysis

Power Production Potential and Carbon Emission Reduction Analysis are vital components to thoroughly assess the feasibility and environmental impact of implementing solar farms on AMLs. The analysis begins by evaluating the solar power production potential of each mine site using geographic data and the PV Watts® Calculator developed by the NREL (NREL 2024). This tool incorporates location including climate conditions, array type (fixed/tracking), and capacity to estimate annual average solar energy generation.

To allow space for on-site infrastructure such as roads, space between panels for maintenance access, inverters, and other necessary equipment, we assume that 50% of the identified area can be covered by solar panels. Given the ongoing advancements in solar cell technology, with panels becoming ever larger and more efficient, this analysis considers two different panel setups. The first case study considers a 500 W PV array facing south with a fixed open rack (mounted on the ground) (NREL 2024). The purpose of this selection was to give a baseline for comparison and to represent current market offerings.

In contrast, the second case study features a 600 W solar panel that's equipped with a 2-axis tracking system to follow the sun's position (NREL 2024). However, it is essential to note that solar panel technology continues to evolve rapidly, with improvements in efficiency and capacity occurring regularly. An average solar panel today could soon become obsolete as advancements continue to push the boundaries of solar energy generation.

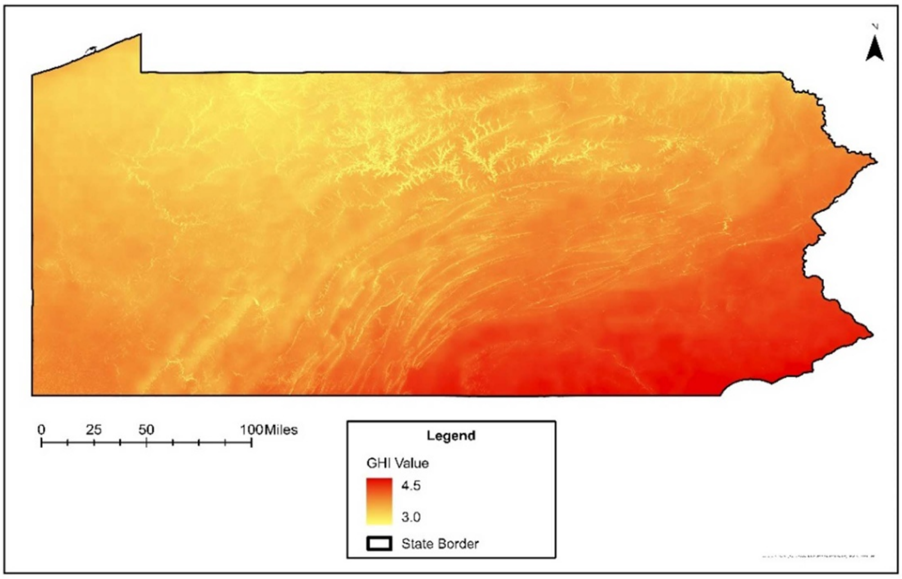

Figure 6. 2019 Global horizontal irradiance of Pennsylvania data source: (Global Solar Atlas 2019).

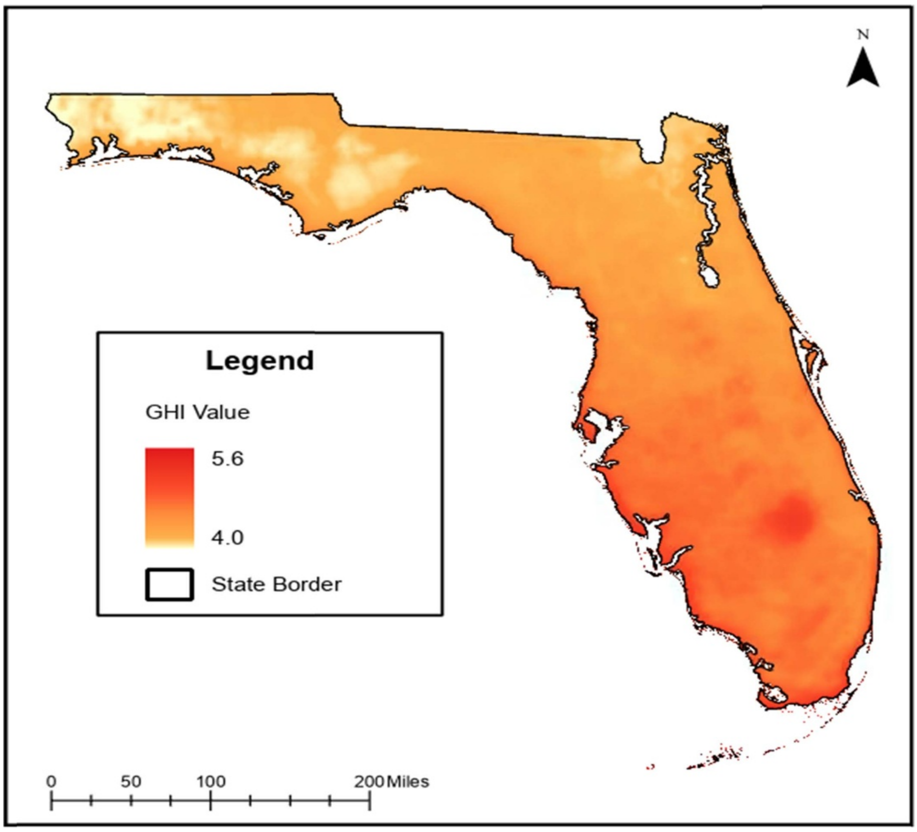

Figure 7. 2019 Global horizontal irradiance of Florida data source: (Global Solar Atlas 2019).

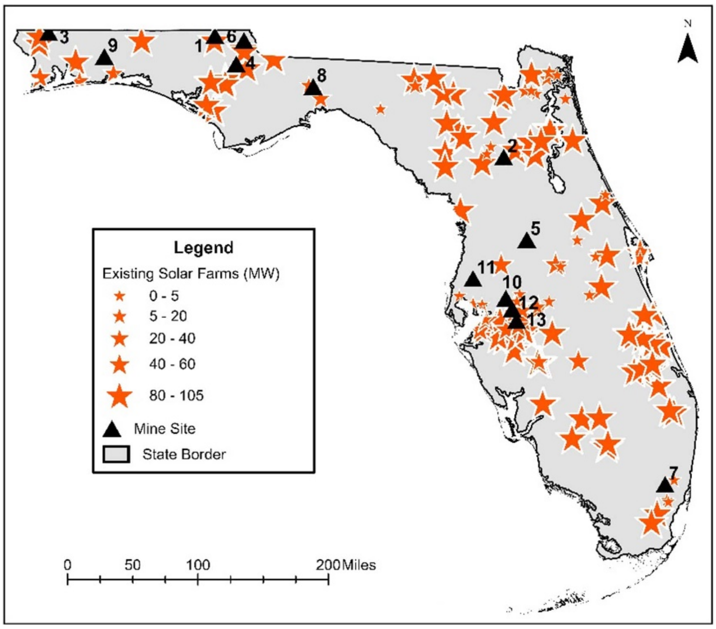

Figure 8. Suitable mine sites and existing solar farms in Florida.

In analyzing the 13 selected sites in Florida, utilizing the specified dimensions and 50% coverage capacity, approximately 44 000 000 solar panels would be installed across all sites. This installation is based on the chosen commercial solar panel dimensions of 78 × 39 inches with 96 solar cells, ensuring uniformity across the study (NREL 2024). Similarly, with the same coverage capacity for Pennsylvania, just over 42 000 solar panels would be installed across suitable mine sites. Notably, five suitable sites were identified in Pennsylvania, contributing to the overall assessment of solar farm development potential on AMLs. These figures demonstrate the extensive scale of solar farm development achievable on AMLs, highlighting the potential for significant RE generation.

The study further calculates the potential to reduce CO_2 emissions using the expected annual solar power production per state multiplied by the amount of CO_2 emitted per MWh from the electricity mix for Florida (97 615 000 metric tons) and Pennsylvania (77 555 000 metric tons) in 2022 (EIA 2023c).

$$\text{MWh x (metric tons CO}_2\text{/MWh)} = \text{metric tons CO}_2 \text{ emissions avoided} \tag{3–2}$$

The equation provided is a formula used to calculate the amount of carbon dioxide (CO_2) emissions avoided based on the generation of electricity in megawatt-hours (MWh) and the corresponding emissions intensity, usually measured in metric tons of CO_2 per MWh. The thorough analysis carried out in this study provides the groundwork for the results section that follows, which will present and discuss in detail the findings pertaining to solar potential, CO_2 emissions reduction, and the environmental impact of solar farm development on AMLs.

3. Results

3.1. Florida

3.1.1. Spatial suitability assessment

The resulting 13 suitable mine sites from the spatial suitability assessment, depicted in figure 8, do not overlay with existing solar farms, ensuring that potential sites for development avoid interference with established RE infrastructure while highlighting the integration potential of repurposed mine lands. However, a range of uncertainty exists in the selection process due to factors such as variability in data resolution and the generalization of land cover classifications. This meticulous filtration process underscores the careful assessment of repurposing AMLs for sustainable energy initiatives across Florida's diverse terrain.

3.1.2. Power production potential and carbon emission reduction analysis

The subsequent analysis delved into the solar potential and environmental impact of the 13 selected sites. Overlaying GHI data and existing solar farms onto these locations confirmed their suitability for solar energy development, validating the efficacy of the methodology. These findings offer valuable insights into Florida's

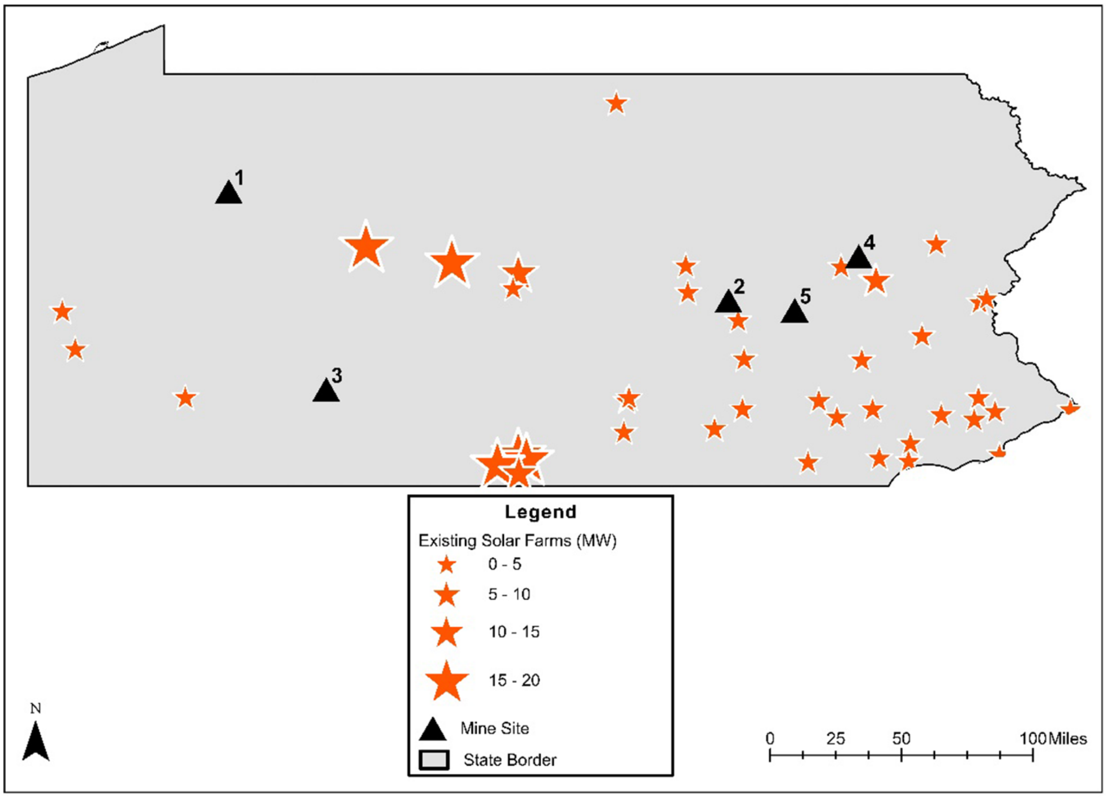

Figure 9. Suitable mine sites and existing solar farms in Pennsylvania.

solar capacity, guiding stakeholders in making informed decisions to advance sustainable energy initiatives tailored to the state's unique geographical and climatic characteristics. In Florida, the flat terrain significantly contributed to the high suitability of selected sites compared to Pennsylvania, where topography posed more challenges.

The 13 suitable sites underwent two case studies for power production potential using the PV Watts calculator. Ranges of estimated potential are provided to reflect variability in solar output due to weather conditions and site-specific factors. The analysis demonstrated significant solar energy potential for the first case study, utilizing fixed open rack panels with a capacity of 500 W each. The 13 sites would generate between 35 000 000 MWh and 40 000 000 MWh of electricity annually, significantly reducing CO_2 emissions by approximately 13 100 000 metric tons. Based on the US average yearly household electricity usage of 10.8 MWh yr^{-1} published by the EIA (EIA 2024) the output powers about 3200 000 houses in Florida. These findings underscore the viability and importance of repurposing AMLs for solar energy production, contributing significantly to Florida's RE goals and environmental sustainability.

The analysis yielded promising results for the second case study with a maximum capacity of 600 W and a 2-axis tracking system PV array. Production estimates range from 55 000 000 MWh to 60 000 000 MWh annually (assuming a $\pm$4% production variability around the mean, based on 30 years of climate data (PV Watts 2024), resulting in approximately 21 600 000 metric tons CO_2 emissions avoided. Considering Florida's (2022) annual electricity generation of approximately 260 000 000 MWh and electricity emissions totaling about 98 000 000 metric tons (EIA 2023c), the 13 sites could facilitate an annual generation equivalent to 17%, leading to a reduction in CO_2 emissions by the same percentage. The output powers approximately 5300 000 homes in Florida, based on US average annual household electricity consumption reported by the EIA (EIA 2024). Please see table 2 for a comprehensive analysis of both case studies summaries of the mine's potential contribution and reduction of carbon emissions. These projections align with the EIA's report on electricity generation and CO_2 emissions from electricity production, affirming the substantial impact of solar energy deployment on AMLs, mitigating environmental impacts, and advancing RE adoption. This analysis underscores the significant potential of repurposing AMLs for solar energy production, contributing significantly to RE goals while mitigating environmental impact.

Table 2. Summation of CO_2 emission reduction and solar power generation potential at selected Florida mine sites.

Case study	Area (m^2)	Solar panels	Installed capacity (MW)	Annual production (MWh)	Households powered annually	% of State's annual electricity production	CO_2 emissions avoided (metric tons)
Case study 1	86 742 724	44 198 755	22 099.4	35 000 000–40 000 000	3200 000	13%–15%	12 800 000–13 500 000
Case Study 2	86 742 724	44 198 755	26 519.3	55 000 000–60 000 000	5300 000	20%–23%	21 000 000–22 000 000

Table 3. Summation of CO2 emission reduction and solar power generation potential at selected Pennsylvania mine sites.

Case study	Area (m^2)	Solar panels	Installed capacity (MW)	Annual production (MWh)	Households powered annually	Percent of state's annual electricity production	CO_2 emissions avoided (tons)
Case study 1	82 708	42 143	21.1	15 000–17 000	1400–1600	0.006%–0.007%	5000–5400
Case study 2	82 708	42 143	25.3	40 000–45 000	3600–4000	0.006%–0.007%	13 000–14 000

3.2. Pennsylvania

3.2.1. Spatial suitability assessment

The five viable mine sites resulting from the spatial suitability assessment are shown in figure 9, where they are strategically overlaid above current solar farms to demonstrate the potential for integration and synergy between converted mine lands and pre-existing RE infrastructure. With a focus on the overall covered area, even with the small quantity, this screening procedure guarantees that only areas with extremely appropriate topography are considered for the construction of solar farms, highlighting the tactical method of choosing locations with the best sun exposure. The primary reason sites in Pennsylvania were deemed unsuitable was due to their slope, which disqualified nearly 70% of the initial pool. Differences in topography compared to Florida made the selection process more restrictive.

3.2.2. Power production potential and carbon emission reduction analysis

The analysis focused on five sites in Pennsylvania, examining their potential for solar development alongside environmental factors. By combining GHI data with maps of existing solar farms, the study validated the effectiveness of the approach and confirmed the suitability of these locations. Additionally, the findings highlighted the distinct position of the selected mine sites, emphasizing their capacity to contribute to Pennsylvania's RE goals. Estimated ranges of production reflect site-specific factors and weather variability. Using the PV Watts calculator, the five suitable sites underwent two case studies for power production potential.

The analysis demonstrated significant solar energy potential for the first case study, utilizing fixed open rack panels with a capacity of 500 W each. The five sites would generate between 15 000 MWh and 17 000 MWh annually, significantly reducing CO_2 emissions to approximately 5200 metric tons. Based on the US average yearly household electricity usage published by the EIA, the output powers about 1500 homes in Pennsylvania. These findings underscore the viability and importance of repurposing AMLs for solar energy production, contributing significantly to Pennsylvania's RE goals and environmental sustainability.

The analysis yielded promising results for the second case study with a maximum capacity of 600 W and a 2-axis tracking system PV array. Production estimates range from 40 000 MWh to 45 000 MWh annually, resulting in approximately 13 400 metric tons of CO_2 emissions avoided. Considering Pennsylvania's annual electricity generation and emissions, the five sites could significantly reduce CO_2 emissions and contribute to the state's RE objectives. The output powers approximately 3800 homes in Pennsylvania, based on the US average annual household electricity consumption. Refer to table 3 for a detailed analysis summarizing the total potential contribution of the mines and the reduction in carbon emissions from the selected sites. These projections align with the state's electricity generation and CO_2 emissions goals, confirming the substantial impact of solar energy deployment on AMLs, mitigating environmental impacts, and advancing RE adoption. This analysis underscores the significant potential of repurposing AMLs for solar energy production, contributing significantly to RE goals while mitigating environmental impact.

4. Discussion

4.1. Clean energy contribution takeaways

The successful execution of the spatial suitability assessments in Florida and Pennsylvania, resulting in the identification of geographically distributed sites for solar farm development, signifies a significant advancement in RE planning. By strategically dispersing the electricity output across various regions, the projects mitigate the risk of overloading the grid in specific areas while maximizing the utilization of renewable resources (Sadat *et al* 2002). The growth of solar capacity in both states, as evidenced by the increasing number of installations and the projected expansion of solar energy production, addresses the demand for battery storage to ensure continuous electricity supply further enhances the reliability and effectiveness of solar energy integration into the grid, demonstrating a tangible commitment to reducing reliance on fossil fuels and transitioning towards cleaner energy sources in both states, as evidenced by increasing installations and projected expansions (DEP 2021, EIA 2023c).

Moreover, the substantial reduction in CO_2 emissions from solar energy deployment on the selected mine sites underscores the pivotal role of repurposing abandoned lands in mitigating environmental impacts and advancing sustainability goals (Olick 2023, The Nature Conservancy 2023). The ability of solar farms on these sites to power thousands of homes signifies a tangible benefit to local communities, not only in terms of reduced energy costs but also in promoting energy independence and resilience (Wiley *et al* 2002).

4.2. Limitations and challenges

It is essential to acknowledge the limitations and challenges encountered during the assessments. In Pennsylvania, the PCA parameters could be further refined to identify more optimal sites, ensuring that areas with improper slopes are not immediately disqualified but assessed for potential development (Pollack 2017). Similarly, in Florida, future analyses could expand beyond AMLs to include other types of brownfields, and reducing reliance on greenfield and agricultural lands (Greenberg *et al* 2001). Additionally, it is crucial to note that solar energy production is inherently limited by its dependence on daylight, necessitating significant investment in battery storage for large-scale development. Furthermore, land ownership, regulatory hurdles, and economic viability pose significant obstacles to solar energy projects and require careful consideration and strategic planning (DEP 2024). The disparity in suitable solar farm sites between Pennsylvania and Florida likely stems from differences in topography, land availability, environmental factors, and regulatory frameworks. Pennsylvania's varied landscape, including mountainous regions and forested areas, has limited the number of viable locations despite its numerous mine sites, while Florida's flatter terrain and potentially more available land contributed to a higher number of suitable sites for solar development.

Addressing these challenges will be crucial for realizing the full potential of solar energy in both states and advancing their clean energy transition agendas. By leveraging federal incentives and programs such as the IRA and the BIL, stakeholders can overcome financial barriers and accelerate the deployment of RE projects (EPA 2023, DOE 2024). Furthermore, state-level initiatives such as the Florida Brownfields Redevelopment Act and Pennsylvania's Brownfield Redevelopment programs offer valuable support for repurposing abandoned lands for sustainable development (FDEP 2024, PADEP 2024).

The successful assessment and identification of suitable sites for solar farm development on AMLs in Florida and Pennsylvania represent significant milestones in advancing RE adoption and environmental sustainability. Stakeholders can capitalize on this momentum by addressing key challenges and leveraging policy support and financial incentives to drive further progress toward a cleaner, more resilient energy future (DEP 2021, EIA 2023c).

4.3. Recommendations for further studies

Further studies could encompass various aspects to enhance the understanding and implementation of solar farm projects on AMLs. Community surveys offer valuable insights into local perspectives and needs, which are crucial for tailoring solar initiatives to specific community requirements. Conducting soil analysis is essential to assess land suitability and identify potential remediation needs. At the same time, a comprehensive Environmental Impact Assessment can provide a holistic evaluation of environmental implications and mitigation strategies. It is vital to consider land limitations which present unpredictable challenges and require thorough investigation from reliable geological sources for effective project planning and risk mitigation. Moreover, integrating long-term climate data that reflects the non-stationary behavior of the atmosphere, and the influence of large scale climate anomalies such as El Nino-Southern oscillation, offers an opportunity to better forecast seasonal variations in production and market electricity need. Future studies could also explore adjusting Pennsylvania's site suitability parameters to expand the pool of optimal locations.

However, it is essential to note that this study did not consider the financial aspects or costs associated with solar panels. Thus, it is recommended that future research explore power purchasing agreements and assess the economic feasibility of solar farm installations on AMLs. Understanding the financial implications and potential return on investment is crucial for stakeholders and decision-makers when evaluating project viability. Future studies could delve deeper into the financial feasibility of solar farm projects to provide a comprehensive analysis of the economic aspects involved, thereby offering valuable insights for sustainable energy development.

5. Conclusion

The urgency of transitioning towards sustainable and RE solutions is underscored by the limitations posed by land availability, a formidable challenge in this era of environmental responsibility. This study has addressed this issue by exploring the option of repurposing AMLs in Florida and Pennsylvania for solar energy production. Against the backdrop of escalating energy demands and the imperative to mitigate climate change, this research provides valuable insights into harnessing innovative solutions to overcome land constraints and accelerate the adoption of RE technologies.

Employing advanced geospatial analysis techniques and integrating diverse datasets, this study has forged a refined and streamlined approach to identifying sites optimal for the reclamation of AMLs. The methodology unveiled in this research delineates a selection of mine sites suitable for solar farm development, establishing a framework for future analyses and initiatives in sustainable energy development. Furthermore, by situating the research within the evolving RE policies of Pennsylvania and Florida, this study underscores the pivotal role of policy frameworks in fostering a cleaner and more sustainable energy landscape.

In conclusion, this thesis marks a significant stride towards unlocking the untapped potential of AMLs for RE development. By marrying clean energy objectives with environmental reclamation, this research offers a pathway for repurposing underutilized land resources and expediting the transition to a more sustainable and environmentally friendly energy future. As nations and states endeavor to meet ambitious clean energy targets, the insights gleaned from this study can serve as a compass for strategic decision-making and policy formulation, propelling progress toward a more sustainable and resilient future. Moreover, researching other site suitability studies can provide valuable comparative insights and bolster the efficacy of such initiatives. These studies can reveal best practices, potential challenges, and innovative solutions, thereby enhancing the overall strategic framework for RE development on AMLs.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary information files).

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